

COMPUTER NETWORKS

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1. Meaning of a Computer Network

A **computer network** is a collection of two or more computers and other digital devices that are connected together so that they can **communicate and share data, resources, and services**.

A network allows computers to exchange information and share resources such as printers, files, applications, and an internet connection.

Example

In a school computer laboratory, 30 computers may be connected to a network. Students can use the network to access shared files, communicate with other computers, and use a common printer.

2. Why Computers Are Connected in a Network

Computers are connected to networks mainly to allow them to **share resources and communicate efficiently**.

A network can be used to share:

- Files and documents
- Printers
- Internet connections
- Software and applications
- Storage resources
- Information
- Messages
- Hardware devices

For example, instead of connecting 20 computers to 20 separate printers, a school can connect all 20 computers to a network and allow them to use one or more shared printers.

3. Basic Components of a Computer Network

A computer network consists of several important components.

A. Computers and Other Devices

These are devices that participate in the network.

Examples include:

- Desktop computers
- Laptops
- Smartphones
- Tablets
- Servers
- Network printers
- Smart TVs
- IP cameras

B. Network Interface Card (NIC)

A **Network Interface Card (NIC)** is a hardware component that allows a computer or other device to connect to a network.

A NIC may provide:

- Wired Ethernet connectivity
- Wireless connectivity

Most modern computers have a network interface built into the motherboard.

C. Network Media

Network media provide the path through which data travels between devices.

They can be:

Wired media

- Twisted-pair cable
- Coaxial cable
- Fibre-optic cable

Wireless media

- Radio waves
 - Microwave
 - Infrared
 - Wi-Fi signals
-

D. Network Devices

Network devices help computers communicate with one another.

Examples include:

- Switch
 - Router
 - Hub
 - Modem
 - Wireless access point
 - Repeater
-

E. Network Software

Network software consists of programs and operating-system features that help devices communicate and share resources.

Examples include:

- Network operating systems
 - Network management software
 - Communication applications
 - Security software
-

4. Types of Computer Networks

Networks can be classified according to the geographical area they cover.

The major types include:

1. PAN

2. LAN
3. MAN
4. WAN

5. Personal Area Network (PAN)

A **Personal Area Network (PAN)** is a small network used to connect devices around an individual. It usually covers a very short distance.

Examples

A smartphone connected to:

- Wireless earbuds
- Smartwatch
- Wireless keyboard
- Wireless mouse

Technologies used

- Bluetooth
- USB
- Personal hotspot

Characteristics

- Covers a small area
- Usually belongs to one person
- Connects personal devices
- Relatively easy to set up

6. Local Area Network (LAN)

A **Local Area Network (LAN)** connects computers and devices within a relatively small geographical area.

Examples include:

- Computer laboratory
- School
- Office
- Home
- Small business

A LAN can use cables, wireless connections, or both.

Example

A school may have a LAN connecting:

- Teachers' computers
- Students' computers
- Server
- Printers
- Administration computers

Advantages of LAN

- Fast communication
- Easy resource sharing

- File sharing
- Printer sharing
- Internet sharing
- Easier management within the organization

7. Metropolitan Area Network (MAN)

A **Metropolitan Area Network (MAN)** connects networks across a larger geographical area, usually covering a city or large town.

It is larger than a LAN but generally smaller than a WAN.

Example

A telecommunications organization may connect several offices located in different parts of the same city.

Characteristics

- Covers a city or metropolitan area
- Connects multiple LANs
- Usually requires more network infrastructure than a LAN

8. Wide Area Network (WAN)

A **Wide Area Network (WAN)** connects computers and networks over a very large geographical area.

It can cover:

- Countries
- Continents
- The entire world

The **Internet** is the world's largest example of a WAN.

Examples

- International company networks
- Bank networks connecting branches across countries
- The Internet

Characteristics

- Covers a very large area
- Connects multiple LANs and other networks
- Can use fibre-optic cables, satellites, microwave links, and other technologies

9. Comparison of PAN, LAN, MAN and WAN

Network	Meaning	Coverage	Example
PAN	Personal Area Network	Very small area	Phone and smartwatch
LAN	Local Area Network	Room, building or campus	School computer lab
MAN	Metropolitan Area Network	City or large town	City-wide network
WAN	Wide Area Network	Countries/continents/world	Internet

A useful way to remember them is:

PAN → LAN → MAN → WAN

from smaller to larger coverage.

10. Network Topology

A **network topology** refers to the physical or logical arrangement of devices and connections in a network.

Common network topologies include:

- Bus
- Star
- Ring
- Mesh
- Tree
- Hybrid

11. Bus Topology

In a **bus topology**, all computers and devices are connected to a single main communication cable called the **backbone**.

Simple representation

Computer — Computer — Computer — Computer

with the devices connected along one main cable.

Advantages

- Simple to install
- Requires less cable
- Relatively inexpensive for small networks

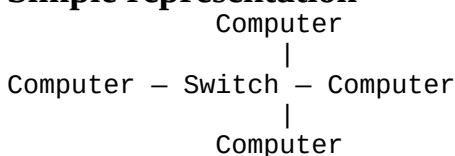
Disadvantages

- Failure of the main cable can affect the entire network
- Performance can decrease when many devices communicate
- Troubleshooting can be difficult
- Limited scalability

12. Star Topology

In a **star topology**, all devices are connected to a central device, usually a **switch**.

Simple representation



Advantages

- Easy to install
- Easy to manage
- Failure of one computer usually does not affect the others
- Easy to add or remove devices
- Faults can be easier to identify

Disadvantage

If the central switch or central device fails, communication between the connected devices may be disrupted.

Star topology is very common in modern LANs.

13. Ring Topology

In a **ring topology**, each device is connected to two other devices, forming a closed loop.

Simple representation

Computer → Computer → Computer → Computer → Computer

with the final computer connected back to the first.

Advantages

- Devices have an organized connection structure
- Data can travel around the ring in a controlled manner

Disadvantages

- Failure of a connection or device can disrupt communication depending on the implementation
 - Adding or removing devices can be more difficult
 - Troubleshooting may be difficult
-

14. Mesh Topology

In a **mesh topology**, devices are connected to many or all other devices.

Types

Full Mesh

Every device has a direct connection to every other device.

Partial Mesh

Some devices have multiple connections while others do not.

Advantages

- High reliability
- Multiple communication paths
- Failure of one connection may not stop communication
- Suitable for networks where reliability is important

Disadvantages

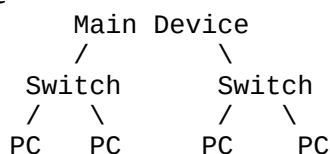
- Expensive
 - Requires many cables or connections
 - Installation and maintenance can be complex
-

15. Tree Topology

A **tree topology** organizes devices in a hierarchical structure.

It can be viewed as a combination of star networks connected in levels.

Example



Advantages

- Easy to expand
- Suitable for large organizations

- Supports hierarchical management

Disadvantages

- Failure of an important higher-level device can affect many devices
 - More complex than a simple star network
-

16. Hybrid Topology

A **hybrid topology** is a network that combines two or more different types of topology. For example, an organization may combine **star and bus topologies**.

Advantages

- Flexible
- Can be designed according to organizational needs
- Suitable for large and complex networks

Disadvantages

- Can be expensive
 - More difficult to design and maintain
 - Requires careful planning
-

17. Network Devices

A. Switch

A **switch** is a networking device used to connect multiple devices within a LAN. It forwards data to the appropriate device based on network addressing information.

Example

A school computer laboratory may use a switch to connect 30 computers.

B. Router

A **router** connects different networks and directs data between them. A router is commonly used to connect a local network to the Internet.

Example

A home router can connect:

- Smartphones
- Laptops
- Smart TVs
- Computers

to the home network and Internet.

C. Hub

A **hub** is a basic networking device that connects multiple devices. Unlike a modern switch, a traditional hub broadcasts incoming data to all connected ports. Hubs are less efficient and are largely replaced by switches in modern Ethernet networks.

D. Modem

The word **modem** comes from **modulator-demodulator**.

A modem is a device that enables digital devices to communicate over certain communication services by converting signals between appropriate forms.

Modems have traditionally been important in connections such as telephone-line Internet access, although many modern broadband devices combine modem and router functions.

E. Wireless Access Point

A **Wireless Access Point (WAP)** provides wireless network connectivity to devices. It allows devices such as laptops and smartphones to connect to a network using Wi-Fi.

F. Repeater

A **repeater** receives and regenerates or retransmits network signals to extend the distance over which communication can occur.

It is useful when a signal needs to travel beyond its normal effective range.

18. Network Transmission Media

Transmission media are the channels through which data travels from one device to another.

They can be divided into:

1. Guided/Wired media
 2. Unguided/Wireless media
-

19. Twisted-Pair Cable

Twisted-pair cable consists of pairs of insulated copper wires twisted together.

Two common types are:

- UTP — Unshielded Twisted Pair
- STP — Shielded Twisted Pair

Twisted-pair cables are widely used for Ethernet networks.

Advantages

- Relatively inexpensive
- Easy to install
- Commonly available
- Suitable for LANs

Disadvantages

- Limited transmission distance compared with fibre
 - Can be affected by electromagnetic interference, especially depending on cable type and environment
-

20. Coaxial Cable

Coaxial cable contains a central conductor surrounded by insulation and shielding.

It has been used in:

- Cable television
- Some networking systems
- Certain communication systems

It provides better shielding than basic unshielded twisted-pair cable.

21. Fibre-Optic Cable

Fibre-optic cable transmits data using **pulses of light through optical fibres**.

Advantages

- Very high data transmission capacity
- High speed
- Can cover long distances
- Resistant to electromagnetic interference
- Difficult to intercept physically compared with ordinary copper cabling

Disadvantages

- More expensive than many copper cables
- Installation can require specialized skills
- Fibre can be physically delicate if not properly protected

Fibre optics are widely used in modern telecommunications and high-speed networks.

22. Wireless Transmission

Wireless networks transmit data without requiring a physical cable between communicating devices.

Examples include:

- Wi-Fi
- Bluetooth
- Microwave
- Satellite communication
- Cellular networks

Advantages

- Reduces the need for cables
- Allows mobility
- Easy to connect portable devices
- Useful where installing cables is difficult

Disadvantages

- Can experience interference
- Wireless signals may have limited range
- Poor security configuration can expose networks to unauthorized access

23. Internet, Intranet and Extranet

Internet

The **Internet** is a worldwide interconnected network of networks that allows devices and networks around the world to communicate.

It provides services such as:

- Websites
 - Email
 - Online communication
 - File transfer
 - Streaming
 - Online learning
-

Intranet

An **intranet** is a private network used within an organization.

For example, a school may have an internal website containing:

- Staff information
- Internal notices
- Student records
- Administrative resources

Access is normally restricted to authorized members of the organization.

Extranet

An **extranet** is a private network or part of an organization's network that provides controlled access to authorized external users.

For example, a company may allow selected suppliers to access certain business information without giving them access to the entire internal network.

24. Network Protocols

A **network protocol** is a set of rules that devices follow when communicating over a network.

Protocols ensure that devices can understand and exchange data correctly.

Common examples include:

TCP/IP

TCP/IP is the fundamental protocol suite used by the Internet and many modern networks.

HTTP

HTTP stands for **Hypertext Transfer Protocol**.

It is used for communication between web browsers and web servers.

HTTPS

HTTPS is the secure version of HTTP. It uses encryption to protect data exchanged between a browser and a website.

FTP

FTP stands for **File Transfer Protocol**.

It is used for transferring files between computers over a network.

SMTP

SMTP stands for **Simple Mail Transfer Protocol**.

It is commonly used for sending email.

DNS

DNS stands for **Domain Name System**.

It helps translate domain names, such as website names, into IP addresses used by networks.

25. IP Address

An **IP address** is a numerical address used to identify a device or network interface on an IP network.

An IPv4 address is commonly written in four numbers separated by periods.

Example:

192.168.1.10

Devices use IP addressing to identify where network data should be sent.

26. MAC Address

A **MAC address** is a hardware-level network identifier associated with a network interface. It is commonly represented as a series of hexadecimal numbers.

Example:

00:1A:2B:3C:4D:5E

A MAC address is different from an IP address.

Simple difference

- **IP address:** used for network-level addressing and routing.
 - **MAC address:** identifies a network interface at the data-link layer.
-

27. Client and Server

Client

A **client** is a device or program that requests a service or resource from another computer.

Examples:

- Web browser
- Email application
- Student's computer requesting a file

Server

A **server** is a computer or software system that provides services or resources to other computers called clients.

Examples:

- Web server
- File server
- Database server
- Email server
- Print server

Example

When a student opens a school's online results system:

Student's browser = Client

Computer providing the website = Server

28. Peer-to-Peer Network

In a **peer-to-peer (P2P) network**, computers can communicate and share resources directly with one another without relying entirely on a dedicated central server.

Advantages

- Simple for small networks
- Can be inexpensive
- Does not necessarily require a dedicated server

Disadvantages

- Difficult to manage as the network grows
- Security can be more difficult to control
- Data management may be less centralized

29. Client-Server Network

In a **client-server network**, clients request services from centralized servers.

Example

A school may have:

- 30 student computers
- 5 teacher computers
- 1 central file server

The server can store files and provide services to authorized users.

Advantages

- Centralized management
- Easier backup
- Better control over user access
- Suitable for larger organizations

Disadvantages

- Server failure can affect services
- Requires administration
- Can be more expensive to set up

30. Advantages of Computer Networks

Computer networks provide many benefits.

1. Resource Sharing

Users can share:

- Printers
- Storage
- Internet connections
- Files

2. Communication

Users can communicate through:

- Email
- Messaging
- Video conferencing
- Voice communication

3. File Sharing

Users can exchange documents and other files.

4. Centralized Management

An administrator can manage users, devices, software, and resources from a central location.

5. Internet Sharing

Several computers can use a common Internet connection.

6. Reduced Costs

Organizations may reduce costs by sharing printers, storage, Internet connections, and other resources.

7. Improved Collaboration

People can work on shared projects and exchange information.

8. Centralized Storage

Important files can be stored on servers and backed up centrally.

31. Disadvantages of Computer Networks

1. Security Risks

Unauthorized users may attempt to access network resources.

2. Malware

Viruses, worms, ransomware, and other malware can spread through networks.

3. Network Failure

A failure of important equipment can interrupt communication and services.

4. Cost

Large networks can require significant costs for:

- Hardware
- Cabling
- Software
- Installation
- Maintenance

5. Technical Skills

Networks require trained people to install, manage, troubleshoot, and secure them.

6. Privacy Risks

Poorly protected networks can expose sensitive information.

32. Network Security

Network security involves protecting a network and its data against unauthorized access, attacks, misuse, damage, or disruption.

Common network security measures include:

Strong Passwords

Users should use strong and unique passwords.

User Authentication

Systems should verify the identity of users before allowing access.

Firewalls

A **firewall** monitors and controls network traffic according to defined security rules.

Antivirus/Anti-malware Software

Security software helps detect and remove malicious software.

Encryption

Encryption converts information into a form that unauthorized people cannot easily understand.

Software Updates

Network devices and software should be updated to fix security vulnerabilities.

Access Control

Users should only have access to resources they are authorized to use.

Backups

Important network data should be backed up regularly.

33. Common Network Threats

Virus

A virus is malicious software that can attach itself to files or programs and spread when the infected file or program is used.

Worm

A worm is malware that can replicate itself and spread across networks without necessarily requiring a user to manually open an infected file.

Trojan Horse

A Trojan is malicious software that disguises itself as legitimate or useful software.

Ransomware

Ransomware is malware that can prevent access to files or systems and demand payment from victims.

Spyware

Spyware is software designed to secretly monitor or collect information about users or their activities.

Phishing

Phishing is a fraudulent attempt to trick people into revealing sensitive information, often through fake messages or websites.

34. Safe Network Practices

Network users should:

- Use strong passwords
 - Never share passwords unnecessarily
 - Avoid suspicious links
 - Avoid downloading files from untrusted sources
 - Keep software updated
 - Use antivirus/anti-malware protection
 - Log out of shared computers
 - Avoid connecting to unknown networks when handling sensitive information
 - Back up important data
 - Report suspicious network activity
-

35. Network Performance

Several factors can affect the performance of a network.

Bandwidth

Bandwidth refers to the amount of data that a network connection can carry over a given period. Higher bandwidth generally allows more data to be transmitted efficiently.

Latency

Latency is the delay between sending data and receiving a response.

Low latency is particularly important for activities such as:

- Video calls
- Online gaming
- Real-time communication

Network Traffic

When many devices send data at the same time, network congestion may occur and performance may decrease.

36. Wired vs Wireless Networks

Wired Network

Uses physical cables
Generally more resistant to radio interference
Often provides stable connections
Requires cable installation
Common in offices and computer labs

Wireless Network

Uses wireless signals
Can be affected by wireless interference
Provides greater mobility
Requires less physical cabling
Common for phones, laptops and tablets

37. LAN vs WAN

LAN

Covers a relatively small area
Usually easier to manage
Generally faster within the local environment
Common in homes, schools and offices
Example: school network

WAN

Covers a very large area
More complex to manage
Performance depends on the connections used
Used to connect networks across large distances
Example: Internet

38. Network Administrator

A **network administrator** is a person responsible for managing and maintaining a computer network.

Their duties may include:

- Installing network equipment
- Configuring network devices
- Managing user accounts
- Monitoring network performance
- Troubleshooting network problems
- Protecting the network
- Managing backups
- Maintaining network documentation

39. Uses of Computer Networks in Everyday Life

Computer networks are used in many areas.

Education

- Online learning

- Sharing educational resources
- School administration
- Online examinations

Banking

- ATM networks
- Online banking
- Electronic transactions
- Communication between bank branches

Healthcare

- Sharing patient information
- Telemedicine
- Hospital management systems

Business

- Email
- File sharing
- Video meetings
- Online sales

Communication

- Messaging
- Voice calls
- Video calls
- Social media

Transportation

- Ticketing systems
- Navigation
- Traffic management
- Airline reservation systems

40. Network Troubleshooting

Network troubleshooting is the process of identifying and fixing network problems.

Common network problems include:

- No Internet connection
- Slow network
- Loose cable
- Incorrect network configuration
- Weak wireless signal
- Failed network device
- Incorrect password
- IP address problems

Basic troubleshooting steps

1. Check whether the device is powered on.
2. Check network cables.
3. Check Wi-Fi settings.
4. Check the network connection.

5. Restart the affected device or network equipment when appropriate.
6. Check network configuration.
7. Test another device.
8. Identify whether the problem affects one device or the entire network.
9. Seek technical assistance if necessary.

41. Summary

A **computer network** is a collection of connected devices that communicate and share resources. Networks can be classified according to size, including **PAN, LAN, MAN, and WAN**. They can also be organized using different topologies such as **star, bus, ring, mesh, tree, and hybrid**. Network devices include **switches, routers, hubs, modems, wireless access points, repeaters, and network interface cards**.

Networks use wired or wireless transmission media and communication protocols such as **TCP/IP, HTTP, HTTPS, FTP, SMTP, and DNS**. Networks provide important benefits such as communication, resource sharing, centralized management, file sharing, and Internet access, but they can also face problems such as security threats, malware, network failures, and high maintenance costs. Proper network security, including **strong passwords, authentication, firewalls, encryption, software updates, access control, and backups**, helps protect network resources and information.

BECE KEY POINTS TO REMEMBER

1. A **computer network** is a group of connected devices that communicate and share resources.
2. **PAN** covers a very small personal area.
3. **LAN** covers a relatively small area such as a school or office.
4. **MAN** covers a city or metropolitan area.
5. **WAN** covers a large geographical area.
6. The **Internet** is the world's largest interconnected network of networks.
7. A **switch** connects devices within a LAN.
8. A **router** connects different networks and directs data between them.
9. A **NIC** allows a device to connect to a network.
10. A **modem** performs signal conversion needed for certain communication links.
11. A **star topology** uses a central connecting device.
12. A **bus topology** uses a main backbone cable.
13. A **ring topology** forms a closed loop.
14. A **mesh topology** provides multiple connections between devices.
15. **Fibre-optic cable** transmits data using light.
16. **Wi-Fi** provides wireless network connectivity.
17. An **IP address** is used for network addressing.
18. A **MAC address** identifies a network interface at the data-link layer.
19. A **protocol** is a set of rules governing communication between networked devices.
20. A **client** requests services, while a **server** provides services.
21. **Network security** protects network resources and data from unauthorized access and attacks.
22. **Firewall, encryption, authentication, strong passwords, updates, and backups** are important network security measures.

23. Networks make **communication and resource sharing** easier.

24. A major disadvantage of networks is the possibility of **security threats and network failure**.

Memorise

Computer Networks: A computer network is a group of two or more computers and digital devices connected together to communicate and share resources such as files, printers, software, storage, and Internet connections. Networks can be classified according to the area they cover: **PAN (Personal Area Network)** for personal devices, **LAN (Local Area Network)** for places such as homes, schools and offices, **MAN (Metropolitan Area Network)** for cities or large towns, and **WAN (Wide Area Network)** for very large areas such as countries and continents. Common network topologies include **bus, star, ring, mesh, tree, and hybrid**. Network devices include **switches, routers, hubs, modems, wireless access points, repeaters, and Network Interface Cards (NICs)**. Data can be transmitted through wired media such as twisted-pair, coaxial and fibre-optic cables, or wirelessly using technologies such as Wi-Fi, Bluetooth, microwave and satellite communication.

Networks use communication rules called **protocols**, including TCP/IP, HTTP, HTTPS, FTP, SMTP and DNS. A **client** requests services while a **server** provides them, and devices can be identified using addresses such as IP and MAC addresses. Computer networks provide benefits such as fast communication, resource sharing, centralized management, collaboration and Internet access, but they can also experience security threats, malware, network failures and maintenance costs. **Network security** protects network resources through measures such as strong passwords, authentication, firewalls, encryption, access control, software updates and regular backups. Networks are widely used in schools, businesses, banks, hospitals, government institutions and communication systems.